

Sampling-Free Oracles for Integration Defects in Quantum Error-Mitigation Pipelines

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Abstract—Quantum error-mitigation benchmarks are software, and an integration error in the pipeline can move their conclusions without producing any error or visibly implausible output. We report a bounded case study of two such errors in a budget-matched zero-noise-extrapolation (ZNE) and readout-error-mitigation (REM) pipeline: unitary folding that rebuilds a transpiled circuit’s terminal measurements can discard the layout-induced qubit-to-clbit map, and readout calibration indexed by virtual rather than physical qubits inverts the response matrix against the wrong qubits. Neither hazard is new – the standard readout-mitigation library provides `final_measurement_mapping` for precisely this purpose, and calibrating the measured physical qubits is documented practice. Our contribution is a reusable validation artifact and a measurement of consequences. We give *two* deterministic, sampling-free oracles, and show they are not interchangeable: the folding oracle ($UU^\dagger U = U$ must leave the ideal distribution unchanged) is provably blind to the calibration defect, because at zero readout noise every response matrix is the identity. On a QAOA MaxCut benchmark over six fixed instances and two device calibration snapshots, run from a fresh seed namespace with paired common random numbers, the defective and corrected implementations differ by 0.001–0.096 of $C_{\max}$ depending on the arm (one arm is unaffected), and the method selected as best changes in 74.7% of cells among the three mitigated methods. That rate overstates the substantive effect: 53.3% of cells are near ties. Selecting on one evaluation sample and paying on an independent one, the defect costs 0.0067 of $C_{\max}$ in excess regret — small in absolute terms, and concentrated on the gate-dominated snapshot. We verify against the standard library used as documented: it agrees with our corrected implementation exactly, and we make no claim that any released library is defective. All results are simulator-only and we make no claim of quantum advantage.

Index Terms—quantum error mitigation, zero-noise extrapolation, readout error mitigation, software correctness, validation, reproducibility, QAOA

I. INTRODUCTION

Comparative benchmarks now guide error-mitigation practice [1]–[3]. Those benchmarks are programs, and their conclusions depend on whether the pipeline that implements the method does what its description says. This paper is a bounded case study of two integration errors that do not announce themselves, together with the oracles that catch them and a measurement of how far they can move a benchmark’s conclusions.

Once a circuit is transpiled to a device coupling map, the transpiler chooses a layout π , so classical bit c records physical qubit $q_{\pi(c)}$. Two things can then go wrong. First, folding that

strips the terminal measurements and rebuilds them with a generic “measure all” operation reinstates $c \leftarrow q_c$ and discards π . Second, readout calibration indexed by virtual qubit inverts the response matrix against the wrong qubits.

What is and is not claimed

Neither hazard is a discovery. Calibrating the physical qubits a circuit actually measures is documented practice, and `mtthree` supplies `final_measurement_mapping` to recover that map from a transpiled circuit [4]. We treat this as established, and our positive control (Section V) shows the library used as documented is unaffected.

We also claim none of the following, all established: that folding leaves state-preparation and measurement error unamplified [5]; that odd integer scale factors avoid partial folds [5]; that circuits should be transpiled before amplification [5]; that shot budget governs which method wins [2]; that accuracy gains need not improve downstream decisions [6]; or that ZNE can show spurious apparent improvement [7].

Contributions

- **Two sampling-free oracles, and a proof they are not interchangeable.** The folding oracle is blind to the calibration defect at zero readout noise; a second oracle with unequal response matrices is required. Both are deterministic and need no noise model (Section IV).
- **A measurement of consequences** on a complete budget-matched benchmark, with instance-level inference and split-sample decision regret (Section VI).
- **A positive control** against an independent, correctly used implementation (Section V).

II. BACKGROUND AND RELATED WORK

ZNE estimates the noiseless expectation value by amplifying noise and extrapolating to zero [8], [9]; digital unitary folding implements the amplification by inserting $U^\dagger U$ pairs [10], as provided by Mitiq [11]. Majumdar *et al.* [5] set out best practices for digital ZNE — transpile before amplifying, prefer odd integer scale factors, and compose with readout mitigation because folding does not amplify SPAM error. We adopt all three as premises.

Readout error is often dominant for shallow circuits [12] and is corrected by inverting a measured response matrix [4], [12]. We use one variant, tensored per-qubit inversion with

clipping; results do not characterise other readout-mitigation methods.

UNITED [2] benchmarks ZNE, Clifford-data regression and virtual distillation against total shots on random circuits and QAOA MaxCut; readout mitigation is not a compared arm. Köster and Mauerer [7] show Richardson ZNE can produce *artefactual* improvement once amplification pushes the signal below the noise floor — a *statistical-regime* phenomenon that occurs in a correct implementation. The errors we study are *integration* errors; they are independent of that regime and are fixed in code. Sampling-cost bounds [13], [14] make cost-equalised comparison the meaningful one.

III. PROBLEM SETTING

Task. QAOA [15] at $p = 1$ for MaxCut on six fixed, pairwise non-isomorphic six-vertex graphs (Table I). For $G = (V, E)$ the objective is $C(z) = \sum_{(i,j) \in E} \frac{1}{2}(1 - z_i z_j)$. With $|V| = 6$ both the exact optimum $C_{\max}$ and the exact noiseless ansatz value are computable, so “error” is unambiguous. The ansatz reference is a multistart numerical best-known value over 60 exact restarts, **not** a proven global optimum, and is labelled as such in the released configuration.

Quantities kept distinct. We separate (i) the difference between two *estimates*; (ii) the difference between two *absolute estimation errors*; (iii) estimator *bias*, the mean signed deviation from the exact value; and (iv) sampling *variance* across seeds. These are reported separately and never used interchangeably.

Noise. Two static IBM calibration snapshots: `dev_lagos` (median readout error 0.169) and `dev_algiers` (0.011). Both `cx-native`, holding the folding primitive fixed. Simulated with Qiskit Aer [16]; **no quantum hardware was used**.

Budget matching. Each estimate receives the same total shots B : the unmitigated arm spends B on one circuit, ZNE $B/3$ at each of scale factors (1, 3, 5), REM $0.8B$ on the circuit and $0.2B$ on $2k$ calibration circuits. Realised totals are asserted per cell; the maximum relative deviation observed was 0.0000%.

Randomness and pairing. All streams derive from a named coordinate hierarchy. The implementation label is *deliberately excluded* from the coordinates, so the defective and corrected runs see identical shot noise for the same cell — common random numbers, which is what makes the implementation contrast a within-cell comparison. Every other factor is a coordinate, so draws are independent across instance, noise, method, budget, seed, basis, scale and replicate. A collision audit over the realised coordinate set reports 0 collisions; we do not claim collisions are impossible.

IV. THE TWO DEFECTS AND THEIR ORACLES

A. Oracle A: folding must preserve the ideal distribution

Global folding at odd scale $2k+1$ implements $U(U^\dagger U)^k = U$, so the *ideal* output distribution must be identical to the unfolded circuit’s. Any difference is a defect, and no noise model is needed. On a four-qubit circuit whose ideal outcome is the single bitstring 0101:

TABLE I
MAXCUT INSTANCES. “BEST $\langle C \rangle$ ” IS A MULTISTART NUMERICAL BEST-KNOWN VALUE, **NOT** A PROVEN GLOBAL ANSATZ OPTIMUM; “AT BEST” IS THE FRACTION OF 60 RESTARTS REACHING IT.

Instance	$ E $	$C_{\max}$	best $\langle C \rangle$	ratio	at best
prism	9	7	5.9392	0.8485	95%
cycle6	6	6	4.5000	0.7500	100%
k33	9	9	6.2321	0.6925	97%
octahedron	12	8	7.3002	0.9125	100%
path6	5	5	3.8679	0.7736	52%
wheel5	10	8	6.3486	0.7936	48%

TABLE II
ORACLE B. MAXIMUM ABSOLUTE ERROR IN THE RECOVERED DISTRIBUTION. THE IDENTITY LAYOUT IS A DEGENERATE CASE IN WHICH THE DEFECT IS INVISIBLE.

initial_layout	aligned calibration	virtual-index calibration
[0, 1, 2, 3]	1.1×10^{-16}	1.1×10^{-16}
[3, 2, 1, 0]	5.6×10^{-17}	1.8×10^{-1}
[1, 3, 0, 2]	5.6×10^{-17}	1.3×10^{-1}
[2, 0, 3, 1]	1.7×10^{-16}	4.9×10^{-2}

initial_layout	defective, scale 3	correct
[0, 1, 2, 3]	0101	0101
[3, 2, 1, 0]	1010	0101
[1, 3, 0, 2]	0011	0101
[2, 0, 3, 1]	1100	0101

The identity layout hides the defect, which is why it survives casual testing. Comparisons are made on *exact probability vectors*, not sampled estimates. For Clifford circuits the agreement is exact to the last bit; with continuous rotations a constant 1.04×10^{-11} offset appears from floating-point angle representation in the inverted gates — measured constant at scales 3, 5 and 9, so it is not length-dependent accumulation, and it is nine orders of magnitude below the smallest effect reported here.

B. Oracle B: calibration must be checked with unequal response matrices

Oracle A cannot detect the calibration defect. It runs at zero readout noise, where every per-qubit response matrix is the identity and any permutation of identities is the identity. We assert this blindness as a test, so the limitation is recorded rather than assumed.

Oracle B uses deliberately unequal response matrices ($p = 0.30, 0.02, 0.15, 0.05$), a non-identity measurement map, and an independently computed reference: exact statevector probabilities propagated through the explicit response matrix. Aligned calibration recovers the ideal distribution to $\sim 10^{-16}$; virtual-index calibration fails by up to 1.8×10^{-1} (Table II).

C. When the defects are invisible

Neither defect is detectable in all circumstances, and we state the conditions rather than claim universality. A non-identity permutation is invisible whenever the state and ob-

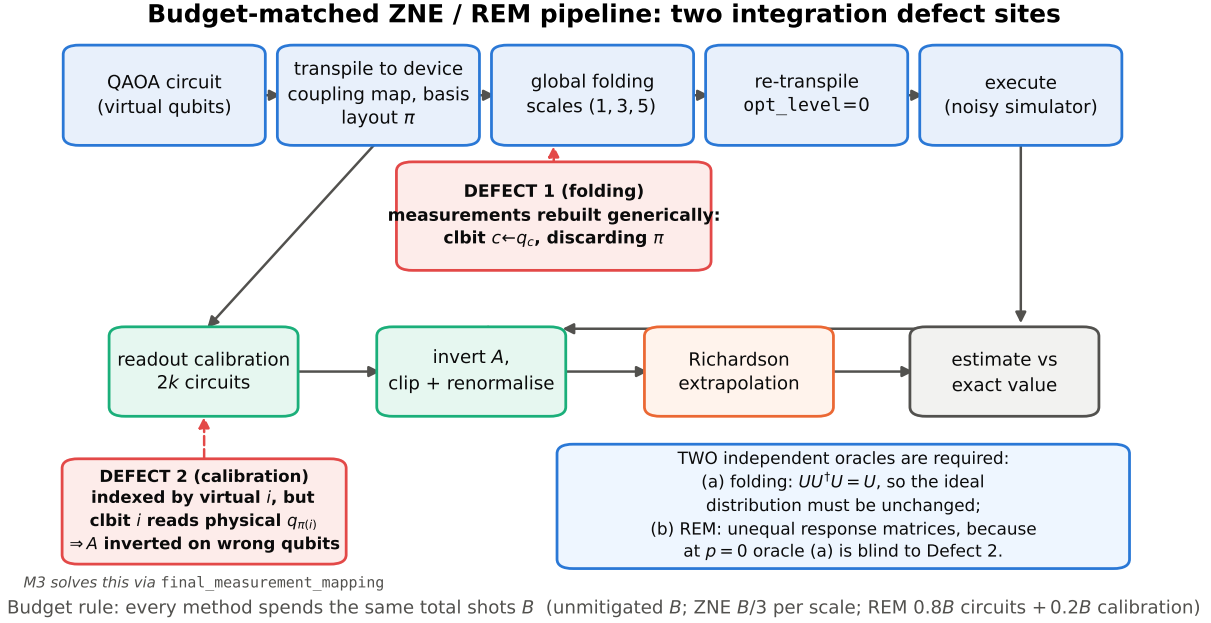


Fig. 1. The budget-matched ZNE/REM pipeline and the two integration defect sites. Both are silent. Two independent oracles are required, because the folding oracle is blind to the calibration defect at zero readout noise.

servable are symmetric under it. Calibration misindexing is invisible whenever the per-qubit response matrices are equal, which includes every zero-readout-noise case.

D. What folding amplifies

Computing exact density-matrix probabilities and applying the readout response matrix explicitly, $p_{\text{meas}} = Ap$: under gate noise alone $\langle ZZ \rangle$ falls from 0.9801 to 0.9044 between scales 1 and 5, whereas under readout noise alone it is 0.8100 at every scale — exactly $(1 - 2p)^2$ for $p = 0.05$. Readout error therefore contributes a *constant attenuation that does not vary with the scale factor*, so folding cannot extrapolate it away. This confirms rather than discovers the point in [5].

V. POSITIVE CONTROL AGAINST AN INDEPENDENT IMPLEMENTATION

We run `mtree` [4] as documented, recovering the measurement map with `final_measurement_mapping`. Its independently implemented map agrees with ours on every layout tested, and both recover the exact distribution to machine precision; only the virtual-index integration fails. **We do not misuse a library and then call it defective**: the contrast is between a correct integration and an incorrect one. Details in the supplementary material.

VI. CONSEQUENCES ON A BENCHMARK

Identical configurations — same instances, noise, budget, seeds — were run through the defective and corrected implementations under common random numbers, from a fresh seed namespace on a clean committed tree. Differences are normalised by each instance’s own C_{max} before averaging.

Inference scope. Primary inference is over the **six fixed instances**, with uncertainty from a stratified bootstrap resampling seeds within each instance. A secondary instance-level cluster bootstrap (6 clusters) indicates what generalisation beyond these instances would look like; it is always wider and is reported alongside.

Table III reports the contrast. The shift is **not** direction-free: directionality exceeds 0.99 for the largest arms, so the defect moves those estimates predominantly one way. Table IV separates estimator bias from sampling variance, which Table III deliberately does not conflate.

Selection changes, with scope stated. Table V reports the rate at which the selected method changes, for the mitigated-only candidate set and for the set including the unmitigated option, together with the near-tie rate. A substantial share of cells are near ties, so a changed winner does not by itself imply a substantive ranking change — which is why we report regret on an independent sample rather than a same-sample minimum. We do not describe any same-sample minimum as “best attainable”.

VII. A PILOT CORRECTION

This work began as a correction to our own unpublished pilot. In that pilot the transpiler chose non-identity layouts for three of twelve conditions, so the mitigated arms there used defective code: 450 of 2400 estimation cells and 30 of 40 optimisation runs. A conclusion the pilot had drawn about in-loop optimisation rested entirely on affected runs and is not carried forward; this paper makes no in-loop claim. We also found, and report, that the pilot’s own “confirmation” run

TABLE III

IMPLEMENTATION CONTRAST, PAIRED WITHIN CELL UNDER COMMON RANDOM NUMBERS, NORMALISED BY $C_{\max}$. *Directionality* IS $|\text{MEAN SIGNED}|/|\text{MEAN}| \cdot |$: VALUES NEAR 1 INDICATE A ONE-DIRECTIONAL SHIFT, NOT CANCELLATION. THE CLUSTER INTERVAL IS THE WEAKER GENERALISATION-ORIENTED INTERVAL. p_{BH} IS ADJUSTED WITHIN THE DECLARED FAMILY; BH ADJUSTMENT DOES NOT REPAIR A SAMPLING MODEL AND IS APPLIED ONLY AFTER THE ESTIMAND AND DEPENDENCE STRUCTURE ARE FIXED.

Noise	Method	n	mean signed	95% CI (stratified)	95% CI (cluster)	mean $ \cdot $	direct.	p_{BH}
dev_algiers	REM	180	+0.0001	[-0.0001, +0.0002]	[-0.0001, +0.0004]	0.0007	0.13	0.293
dev_algiers	ZNE	180	+0.0623	[+0.0597, +0.0649]	[+0.0328, +0.0971]	0.0630	0.99	3.7×10^{-30}
dev_algiers	ZNE+REM	180	+0.0728	[+0.0698, +0.0757]	[+0.0396, +0.1119]	0.0732	0.99	3.5×10^{-30}
dev_lagos	REM	180	-0.0377	[-0.0466, -0.0283]	[-0.0483, -0.0297]	0.0594	0.64	2.8×10^{-14}
dev_lagos	ZNE	180	-0.0211	[-0.0222, -0.0200]	[-0.0357, -0.0092]	0.0218	0.97	2.9×10^{-28}
dev_lagos	ZNE+REM	180	-0.0581	[-0.0726, -0.0432]	[-0.0762, -0.0389]	0.0957	0.61	3.0×10^{-13}

TABLE IV

ESTIMATOR BIAS AND SAMPLING VARIANCE, REPORTED *separately* FROM DIFFERENCES OF ABSOLUTE ERRORS AND NORMALISED BY $C_{\max}$. NOTE THAT ON DEV_ALGIERS THE DEFECTIVE IMPLEMENTATION GIVES ZNE A LARGE *positive* BIAS WHERE THE CORRECTED ONE IS NEAR ZERO — A ONE-DIRECTIONAL SHIFT, NOT A SYMMETRIC SCATTER.

Noise	Method	Impl.	bias	95% CI	sd	RMSE
dev_algiers	unmitigated	v2	-0.0297	[-0.0300, -0.0294]	0.0021	0.0299
dev_algiers	REM	legacy	-0.0241	[-0.0246, -0.0237]	0.0032	0.0245
dev_algiers	REM	v2	-0.0240	[-0.0245, -0.0236]	0.0033	0.0244
dev_algiers	ZNE	legacy	+0.0714	[+0.0691, +0.0737]	0.0165	0.0842
dev_algiers	ZNE	v2	-0.0064	[-0.0078, -0.0050]	0.0096	0.0116
dev_algiers	ZNE+REM	legacy	+0.0807	[+0.0781, +0.0835]	0.0188	0.0947
dev_algiers	ZNE+REM	v2	-0.0011	[-0.0026, +0.0004]	0.0106	0.0105
dev_lagos	unmitigated	v2	-0.1467	[-0.1471, -0.1463]	0.0030	0.1494
dev_lagos	REM	legacy	-0.0953	[-0.1033, -0.0878]	0.0533	0.1107
dev_lagos	REM	v2	-0.1330	[-0.1385, -0.1279]	0.0365	0.1393
dev_lagos	ZNE	legacy	-0.1092	[-0.1107, -0.1076]	0.0109	0.1110
dev_lagos	ZNE	v2	-0.1303	[-0.1317, -0.1288]	0.0101	0.1344
dev_lagos	ZNE+REM	legacy	-0.0644	[-0.0785, -0.0511]	0.0953	0.1153
dev_lagos	ZNE+REM	v2	-0.1373	[-0.1466, -0.1282]	0.0636	0.1532

TABLE V

METHOD SELECTION. TWO CANDIDATE SCOPES ARE REPORTED: THE THREE MITIGATED METHODS, AND THE SAME SET WITH THE UNMITIGATED OPTION ADDED. *Near-tie* IS THE FRACTION OF CELLS WHOSE BEST AND SECOND-BEST CANDIDATES DIFFER BY LESS THAN $0.01 C_{\max}$ — IT IS LARGE, SO A CHANGED WINNER OFTEN REFLECTS A NEAR-TIE RATHER THAN A SUBSTANTIVE RANKING CHANGE. *Split-sample regret* SELECTS THE METHOD ON ONE EVALUATION REPLICATE AND MEASURES THE CONSEQUENCE ON AN INDEPENDENT ONE; THE SAME-SAMPLE MINIMUM IS NEVER DESCRIBED AS “BEST ATTAINABLE”.

Noise	n	change (3 mitig.)	change (incl. unmit.)	near-tie	regret defective	regret corrected	excess
dev_algiers	180	91.1%	91.1%	78.9%	0.0162	0.0039	+0.0123
dev_lagos	180	58.3%	58.3%	27.8%	0.0290	0.0280	+0.0011
all	360	74.7%	74.7%	53.3%	0.0226	0.0159	+0.0067

was not independent: its seed coordinates omitted the instance, noise, method and budget, so all 224 of its rows reproduced main-study values bit-for-bit. That pilot data is retained as a single exploratory campaign and no confirmatory claim rests on it. We record this because it is the reason the oracles in Section IV exist, not as a contribution.

VIII. THREATS TO VALIDITY

Simulator only. Aer snapshots omit crosstalk, drift, leakage and non-Markovian effects. **Scope.** Six fixed instances, $p = 1$, six qubits; instances are not a random sample and generalisation beyond them is indicated only by the wider cluster interval. **Single implementations.** Global folding, Richardson

extrapolation, tensored inversion with clipping; results do not characterise other readout-mitigation methods or local folding. **Device family.** IBM `cx`-native snapshots only. **Generality of the defects.** Demonstrated in one pipeline built on widely used components. We have not audited third-party libraries, and the positive control indicates the standard library used as documented is unaffected. **Dependence.** Seeds are nested within instance and recur across arms; observations are not independent draws and are not analysed as such. **Round-1 data** was generated with correlated streams and a dirty working tree; it is reported as exploratory, its provenance is reconstructed rather than relabelled, and no claim rests on it.

Implementation contrast on fresh confirmation data (simulator only)

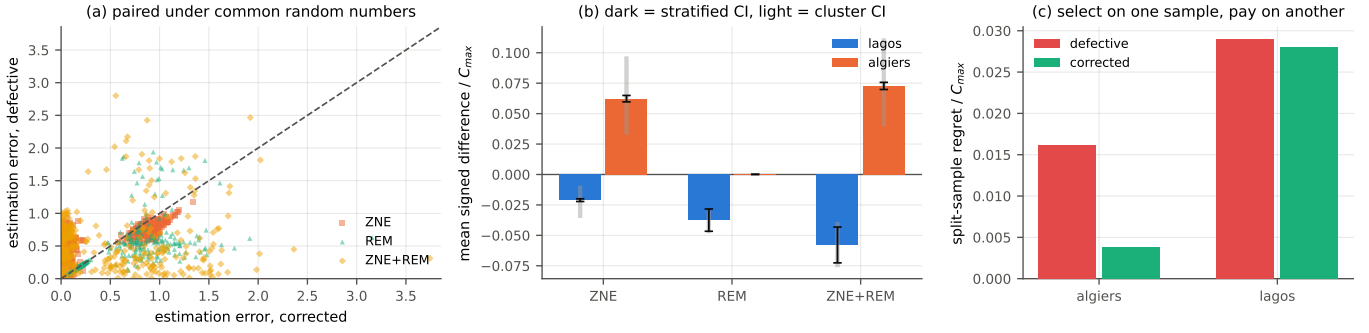


Fig. 2. Implementation contrast on the fresh confirmation data. (a) Per-cell estimation error, defective versus corrected, paired under common random numbers. (b) Mean normalised difference with stratified (solid) and cluster (light) intervals. (c) Split-sample regret. Simulator only.

Per-instance results, corrected implementation (simulator only; 30 seeds per bar)

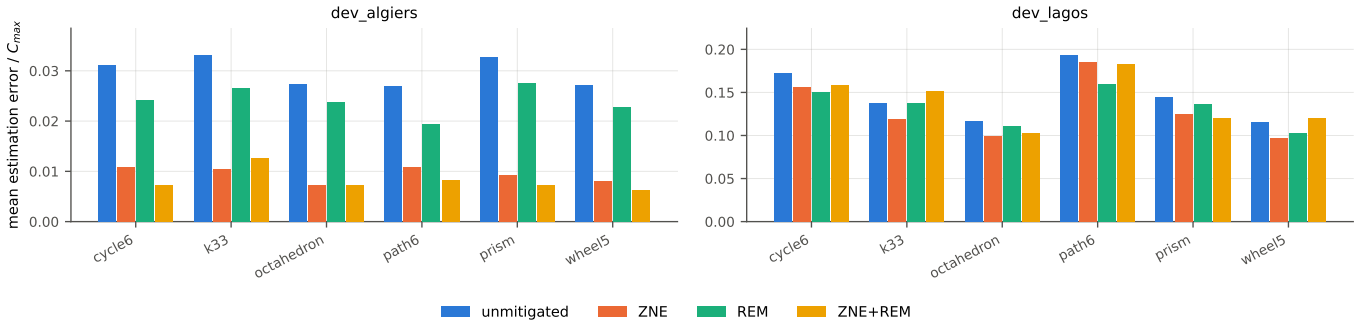


Fig. 3. Per-instance results. Pooled means conceal instance-to-instance heterogeneity, which is why inference is instance-aware. Simulator only.

IX. REPRODUCIBILITY AND DATA AVAILABILITY

Code, protocols, configurations, raw per-cell records, processed results, figure sources and the regression tests accompany this submission with a pinned environment. Every number is generated from raw data by the analysis scripts; none is transcribed by hand, and a traceability table maps each quantitative claim to a file, configuration, command and output table. Round-2 rows record the SHA-256 content hash of every execution-path file. The repository is currently private and will be made public on acceptance; we do not claim public availability before then.

X. CONCLUSION

Two integration errors in an error-mitigation pipeline changed which method a benchmark selected, without producing any error or implausible output. Neither hazard is new, and the standard tooling already solves one of them. What we offer is a pair of deterministic, sampling-free oracles — and the observation that they are not interchangeable, since the folding oracle is provably blind to the calibration defect — together with a measurement of how far such an error can move a complete benchmark. We recommend asserting

the measurement map after any circuit rewrite, and indexing readout calibration by the physical qubits the classical bits actually read.

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